



OPEN Network intrusion detection model using wrapper based feature selection and multi head attention transformers

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Nowadays, many fields, such as healthcare, farming, factories, transportation, cities, and homes are connected via network devices. These systems are configured in open environments and are prone to malicious attacks. It is important to protect these systems from intruders and cyberattacks. Due to the increase in data, the diverse nature of devices, and the types of attacks, standard security systems find it difficult to tackle these attacks. Many researchers have worked to address the problem of intrusion detection in networks. Machine learning and deep learning have also been used. Despite the strong literature, the accuracy of the methods is still an open issue. This article presents a model for intrusion detection with improved accuracy using the UNSW-NB15 dataset. The model uses a wrapper-based feature selection technique using machine learning algorithms to select the best features, which are then combined and fed into a Multi-Head Attention-based transformer for getting the predictions. Accuracy, Precision, recall, and F-1 score are used to evaluate the model. The proposed model improves the accuracy of intrusion detection by selecting the most relevant features while reducing the feature space. The results are compared with other state-of-the-art methods to prove the validity of the proposed model.

Keywords Network security, Intrusion detection, Deep learning, Multi-head attention transformer, Wrapper-based feature selection

Computer networks are used in many fields in different forms, i.e., wired, wireless, Internet of Things (IoT), etc. The domains include healthcare, farming, homes, cities, and transportation, etc. These technologies connect machines in factories, manage buildings, make transportation systems smarter, analyze data, monitor health, and more¹. Hackers might break into systems to steal private information, mess with the money, or spy on them. The risks associated with these technologies make it tough for the industry to grow and be used everywhere. Keeping networks and devices secure is important for researchers². Various studies have been conducted on the detection of intrusion, which is seen to be the most effective approach for solving these issues.

Intrusion detection systems are divided into three main types based on how they identify threats named as anomaly-based, specification-based, and signature-based³. As new network attacks have emerged in recent years and due to the rise in network data traffic, network intrusion detection has become more difficult. Machine learning has become a dominant tool in these systems because of the learning capabilities from complex input using advanced statistical methods and algorithms⁴. Machine learning-based intrusion detection systems are divided into supervised and unsupervised learning⁵. Deep learning, a different discipline, depends on neural networks with two or more hidden layers. Figure 1 shows the taxonomy of intrusion detection systems.

Many researchers have worked on developing intrusion detection systems using machine learning algorithms^{6–8}. Deep learning has also diverse applications in such problems. These algorithms are best at handling high-dimensional data and complicated features compared with traditional machine learning algorithms. Many researchers have applied deep learning for intrusion detection^{9–12}. The detection systems are struggling to cope with modern challenges and the increasing complexities of attacks. Methods that improve the accuracy of anomaly detection by utilizing reduced features and advanced deep learning techniques are needed. This article

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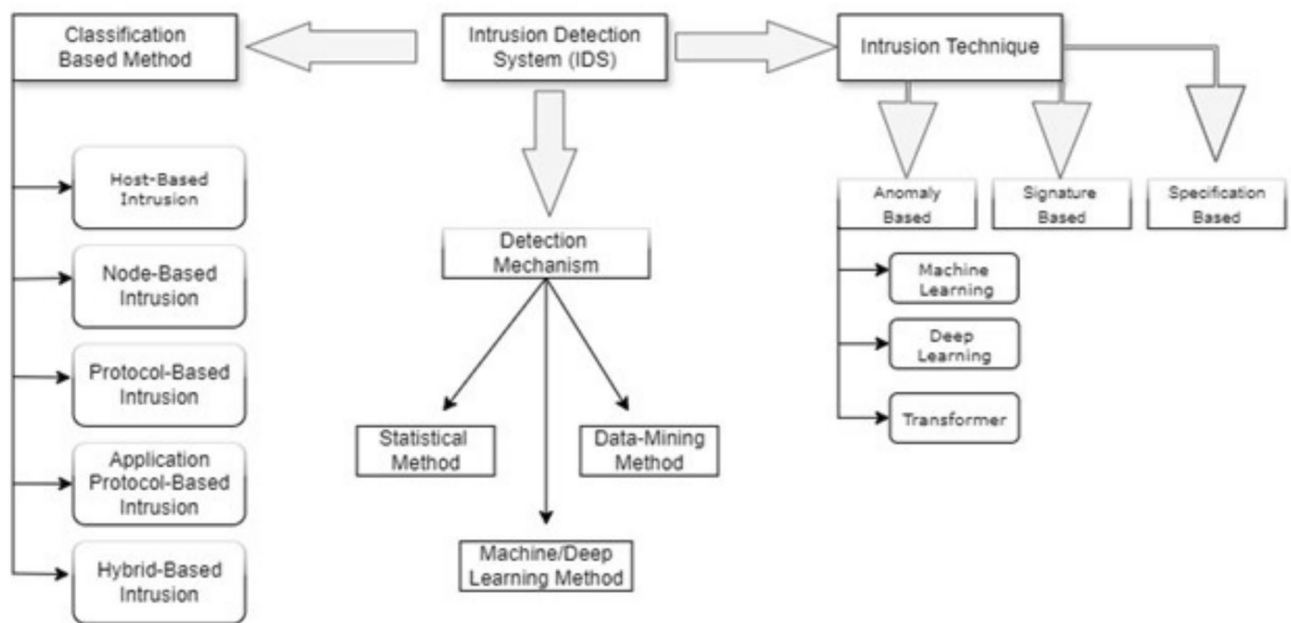


Fig. 1. Taxonomy of intrusion detection systems.

focuses on the analysis of data-based strategies because these methods are fast and capable of handling unknown attacks. The proposed model first uses a wrapper-based method to select and extract essential features. These features are then passed on to the encoder-based transformer model, which predicts whether the next record is an attack or a normal record. The motivation for using Transformers in intrusion detection systems is the ability to capture relationships between features through the self-attention mechanism. The proposed work is evaluated using the UNSW-NB15 dataset¹³ using benchmark metrics. The results are compared with other models to validate the performance of the proposed model.

The remainder of the paper is organized as follows. Section 2 presents literature review that covers state-of-the-art methods using machine learning, deep learning, and transformers. Comparative analysis of different methods is presented. Section 3 presents materials and methods followed by results and discussion in Section 4. Section 5 concludes the article with future research directions.

Literature review

Machine learning and AI have been integrated into various aspects of cybersecurity across multiple sectors, including the detection of intrusions and malware. This section presents a thorough analysis of the literature using machine learning, deep learning, and transformers for intrusion detection. Comparative analysis of the relevant methods is presented. For further details on intrusion detection the readers are referred to review articles that cover different methods^{14–16}.

Machine learning-based intrusion detection

These approaches have gained popularity in intrusion detection in recent years. These methods can analyze large datasets to identify features and patterns related to intrusions, which helps to improve both detection accuracy and speed. Fuzzy C-Means clustering algorithm is applied to six different parts of the KDD Cup 1999 dataset, each containing 5,000 records in¹⁷. The detection rate ranged from 50.3% to 88.5%, with a false positive rate between 0.2% and 4.1%. An improved K-Means algorithm to address sensitivity issues related to the selection of initial centers is presented in¹⁸. The algorithm was tested using the KDD Cup 1999 dataset. K-Means clustering technique is widely used for its simplicity in various applications, including intrusion detection. Min-Max K-Means, a modified K-Means method is presented in¹⁹. The experimental results show a good detection rate as compared to the regular K-Means algorithm, with a lower false positive rate. Another study uses clustering techniques to reduce the high false positive rate in intrusion detection systems²⁰. The authors used the DARPA 2000 and LLDOS datasets to validate the experiments with the K-Means and fuzzy C-Means approaches. The results show that the Fuzzy C-Means algorithm gives good results than the K-Means. In another study, the score of various models was implemented using Apache Spark in²¹. Results are compared with other works utilizing the UNSW-NB15 dataset. Random Forest (RF) model outperformed others, providing the best accuracy compared to Decision Tree (DT) and Naïve Bayes (NB) classifiers. A machine learning-based algorithm for intrusion detection is presented in²². The catBoost technique is used to achieve the desired goal. The NSL-KDD dataset is used to evaluate the proposed model using standard parameters. A machine learning ensemble-based method for detecting network intrusions is proposed in⁸. The model used the UNSW-NB15 dataset for validation and achieved 92.10% accuracy. Utilizing the Synthetic Minority Oversampling Technique (SMOTE), the issue of data imbalance was resolved. Using Principal Component Analysis (PCA), closely related features were selected,

and the RF technique was utilized for prediction. The analysis in²³ evaluated the Classification and Regression Trees (CART) model using a hybrid approach that combined RF for feature selection with CART for classifying different attack types. This method achieved better performance with improved accuracy. UNSW-NB15 and CICIDS2017 datasets are used for intrusion detection in²⁴. The results conclude that using Support Vector Machine (SVM) with feature embedding via NB resulted in better accuracy. To improve the detection accuracy of the intrusion detection system, an ensemble machine learning technique is proposed²⁵. AdaBoost, RF, and Logistic Regression (LR) are used to integrate the voting ensemble model. The proposed model is evaluated with the KDDCUP99 dataset. Another model based on based on the voting ensemble method is presented in²⁶. The proposed model uses two methods i.e. SVM and ExtraTree. KDDCup99 dataset is used to validate the proposed model using accuracy as the main parameter. To reduce misclassification errors in DDoS detection, a model is proposed that uses two feature selection methods to select more relevant features²⁷. The methods include Mutual Information (MI) and Random Forest Feature Importance (RFFI). Different machine learning models are applied to selected features. The CICIDS 2017 dataset is used to evaluate the proposed model. In another study the authors work on stacking machine learning models with Mutual Information Gain and Extra Tree Classifier for selection of attributes²⁸. SVM-based model for intrusion detection is proposed in⁷. Using the UNSW-NB15 dataset, this approach achieved better detection rate; however, the model is unable to detect newly launched attacks.

Table 1 summarizes different methods for intrusion detection using machine learning.

Deep learning-based intrusion detection

The popularity of these methods in intrusion detection has significantly increased in recent years. Unlike traditional machine learning algorithms, deep learning is more effective in managing high-dimensional data and handling complex features. A survey on intrusion detection systems in IoT domain analyzed systems based on various factors, including identification methods, system location, validation techniques, and methodologies for ensuring protection against attacks²⁹. In a study, DT and Genetic Algorithms (GA) are integrated for intrusion detection, and validated on various datasets³⁰. Similarly, in another study³¹ a system for detecting and preventing DoS attacks by integrating intrusion detection is presented. Another technique used deep learning and ensemble techniques in network intrusion detection³². Three approaches were compared with a single Long Short-Term Memory (LSTM) model, a homogeneous ensemble using a tuned RF classifier, and a heterogeneous ensemble combining multiple classifiers. Using the UNSW-NB15 dataset, the LSTM approach attained detection accuracies of 80% and 72% for two-class and more than two-class classifications, respectively. The homogeneous ensemble significantly improved performance, achieving accuracies of 91% for two-class and 87.4% for more than two-class classification. In another study³³ the authors proposed a system to identify botnets using Convolutional Neural Network (CNN) with one-class classification. They introduced a hybrid deep learning model that integrated CNN with Weight-Dropped Long Short-Term Memory (WDLSTM). The CNN extracts essential attributes from the intrusion detection system, while the WDLSTM addresses long-lasting dependencies and gradient vanishing issues. The UNSW-NB15 dataset was used for analysis, achieving better accuracy. A Bi-LSTM-RNN technique to detect intrusions³⁴, differentiating between normal and abnormal activities. The model was trained on the UNSW-NB15 dataset and achieved better accuracy. Feature Extraction Unit (WFEU) based on wrappers is presented in³⁵. The method produced an optimized attribute vector for the Feed-Forward Deep Neural Network (FFDNN) for Wi-Fi intrusion detection using the extra trees technique. The performance was assessed on UNSW-NB15 and AWID datasets and compared against other machine learning algorithms such as RF, SVM, NB, and DT. In another article³⁶, the authors proposed a statistical intrusion detection system to address recent attacks such as DoS attacks, probes, and other malicious activities. Using the UNSW-NB15 dataset, the system applied feature selection via Information Gain (IG), reducing features from 47 to 13. The classification algorithm achieved a score of 92.37%. Authors in³⁷ proposed a hybrid intrusion detection method that utilized SMOTE and Operational Support Systems (OSS) for class balancing. Using the NSL-KDD and UNSW-NB15 datasets, the CNN model achieved better results, scoring 83.60% on NSL-KDD and 78.10% on UNSW-NB15. A combined LSTM and CNN strategy was proposed to enhance intrusion detection accuracy, particularly for unbalanced datasets in³⁸. The authors employed a hybrid technique called STL, which combined SMOTE with Tomek Links

Technique	Dataset Used	Weaknesses
Fuzzy C-Means clustering ¹⁷	KDD Cup 1999	Accuracy of base classifier needs to be improved
Enhanced K-Means ¹⁸	KDD Cup 1999	Sensitivity to selection of initial centers
Min-Max K-Means ¹⁹	UNSW-NB15, KDD Cup 1999	Unable to detect unknown attacks
K-Means, Fuzzy C-Means ²⁰	DARPA 2000, LLDOS	High false positive rate with K-Means; Fuzzy C-Means performs better
RF,DT,NB ²¹	UNSW-NB15	Low detection accuracy
Stacking machine learning models with Mutual Information Gain and Extra Tree Classifier ²⁸	UNSW-NB15	Low detection accuracy
SVM-based model ⁷	UNSW-NB15	Unable to detect newly launched attacks
Machine learning ensemble method (RF) ⁸	UNSW-NB15	Accuracy needs improvement; SMOTE may lead to noisy data and overfitting

Table 1. Comparative analysis of machine learning based intrusion detection methods.

to create a balanced training dataset. The approach was evaluated using CICIDS-001 for multi-class classification and UNSW-NB15 for binary classification, achieving 88.83% for multi-class and 91.17% for binary classification. Authors in³⁹ studied the effectiveness of using various deep learning models for intrusion detection using the UNSW-NB15 dataset. The study compared Artificial Neural Networks (ANN), Recurrent Neural Networks (RNN), and Deep Neural Networks (DNN). Data preprocessing included addressing missing values, category transformations, and min-max normalization. The ANNs achieved the highest scores of 91.26% for two-class classification and 91.89% for more than two-class classification. DNNs followed closely, with scores of 93.22% and 92.90%, respectively, while RNNs had the lowest performance at 85.42% and 85.40%. The effectiveness of DNN-based intrusion detection was further investigated in⁷. The model was tested on three standard datasets including UNSW-NB15, KDDCUP'99, and NSL-KDD, achieving a consistent accuracy of 91.50%. This demonstrated the potential of DNNs for developing robust intrusion detection systems. In another study, the authors used hybrid deep learning model for DDoS attack detection in IoT networks⁴⁰. Various deep-learning models are used for feature extraction and attack detection. The CICIOT2023 dataset is used for evaluation that shows better results in terms of accuracy.

Table 2 shows an overview of different intrusion detection systems using deep learning algorithms.

Transformer-based intrusion detection

Intrusion detection methods using machine learning and deep learning have given promising results in the past. However, challenges such as longer training times, data balancing issues, and suboptimal performance in multi-class classification still exist. To address these issues, researchers have utilized the transformers in intrusion detection. Using position embedding to capture sequence information across attributes and stacked encoder-decoder variants of the transformer to generate low-dimensional representation of features, the authors in⁴¹ proposed a Robust Transformation-based Intrusion Detection System (RTIDS). This system effectively characterizes various types of network traffic using self-attention mechanism. However, the ability of the encoder to generate low-dimensional representations was reduced once this information passed through the decoder. A sliding window mechanism combined with an enhanced visual transformer model for intrusion detection is proposed in⁴². This model improves local attribute modeling using a hierarchical focal loss function to address classification and imbalanced dataset issues. Another hybrid model combining CNNs and transformers is proposed in⁴³. The model aims to recognize intrusions and learn global and local packet correlations but lacks significant contributions to multi-class classification. Authors in⁴⁴ present a model called BBO-CFAT that combines the biogeography-based optimization algorithm for selecting the best attributes with a modified transformer to keep contextual information and reduce complexity. BBO-CFAT applies the roulette selection method for migration and mutation operations, alongside feature information entropy, to improve the reliability of the feature selection. The hierarchical structure of the model captures more contextual information using depth-wise separable convolutions to improve computational efficiency. Experiments were conducted on the CIC-IDS2017 and NSL-KDD datasets. Vaswani et al.⁴⁵ demonstrated improved scores in parallel training, surpassing other models giving better results with reduced training time. This has sparked interest in utilizing transformers within the domain of intrusion detection. An improved transformer for network intrusion detection is presented in⁴⁶. The approach utilizes a stacked autoencoder for dimensionality reduction and a hybrid data sampling technique combining KNN-based undersampling with borderline-SMOTE to address class imbalance issues. Enhanced position encoding in the transformer improved classification performance. The model employed a stepwise learning process, providing binary predictions of intrusion events before moving to multi-class predictions. Testing was conducted on the NSL-KDD dataset. A hybrid attribute selection method, IGRF-RFE, for more than two-class network anomaly detection using a multi-layer perceptron (MLP) is proposed in⁴⁷. The approach combines the speed of filter-based techniques with the relevance of wrapper methods. In the first phase, information gain and RF reduce the feature subset by removing less important features. In the second phase, Recursive Feature Elimination (RFE) refines the reduced features set. Results on the UNSW-NB15 dataset show that IGRF-RFE reduces feature dimensions from 42 to 23 and improves classification accuracy. Authors in⁴⁸ present a Dilated Convolutional Transformer-based GAN (DCT-GAN) designed to improve training, accuracy, and generalization. DCT-GAN employs multi-generators with a single classifier to prevent mode collapse. Each generator utilizes a dilated CNN combined with a transformer block to extract fine-grained and coarse-grained information for better generalization. Experimental results confirmed the effectiveness of DCT-GAN and its components. Authors in⁴⁹ proposed the IDS-INT method to detect various types of attacks in imbalanced network traffic. IDS-INT applies transfer learning to understand feature relationships under imbalanced data conditions. SMOTE is used to address the minority class imbalance problem, followed by feature extraction through a CNN model. Finally, a CNN-LSTM model detects different

Technique	Dataset Used	Weaknesses
AlexNet ³⁰	NSL-KDD, UNSW-NB15	Low detection accuracy
Integration of Intrusion Detection and Prevention Systems ³¹	NSL-KDD, UNSW-NB15	Falls short in identifying attacks other than DoS
LSTM, RF ³²	UNSW-NB15	High false positive rate
Bi-LSTM-RNN ³⁴	NSL-KDD	Complex computational resources are needed
LSTM, CNN with STL ³⁸	CICIDS-001, UNSW-NB15	Complex computational resources are needed; Low accuracy
ANN, RNN, DNN ³⁹	UNSW-NB15	Accuracy of some models needs to be improved

Table 2. Comparative analysis of deep learning based Intrusion detection methods.

attack types. In another study⁵⁰ the authors introduced a new intrusion detection approach by combining transformers and bidirectional long short-term memory (BiLSTM). This method uses a transformer encoder and BiLSTM for intrusion detection, with deep neural networks for feature extraction. The NSL-KDD dataset was used to validate the method's effectiveness.

Table 3 presents a comparative analysis of different techniques for intrusion detection using transformers.

Materials and methods

This section discusses the proposed approach for detecting network intrusion. The aim is to develop a robust methodology for detecting attacks in computer networks utilizing machine learning techniques for feature selection and transformers for attack detection. The proposed methodology is focused on optimizing the process of selecting features to increase the performance of transformer-based models in detecting various categories of network attacks. The work follows an experimental design, which involves the collection of network traffic data, pre-processing, feature selection, model training, and evaluation. The experimental setup is divided into several phases. Figure 2 and Algorithm 1 illustrate the working of the proposed model. The following subsections present details of each step.

Input: Raw network traffic dataset D

Output: Predicted intrusion class labels $\hat{Y}$

Phase 1: Data Preprocessing

Clean and prepare raw dataset D ;

Convert categorical features to numerical vectors using Word2Vec $\rightarrow D_{\text{encoded}}$;

Detect and remove outliers using Z-Score method $\rightarrow D_{\text{clean}}$;

Normalize numeric attributes using Min-Max scaling $\rightarrow D_{\text{norm}}$;

Phase 2: Class Balancing

Apply SMOTE to handle class imbalance and synthesize minority class samples $\rightarrow D_{\text{balanced}}$;

Phase 3: Wrapper-Based Feature Selection

Initialize machine learning models: SVM, Random Forest (RF), and Naïve Bayes (NB) ;

foreach model M in {SVM, RF, NB} **do**

 Rank features based on model-specific importance using cross-validation ;

Aggregate top-ranked features from all models to form optimal feature subset F_{selected} ;

Phase 4: Model Development with Transformer

Split $D_{\text{balanced}}[F_{\text{selected}}]$ into training, validation, and testing sets ;

Build Transformer encoder with the following layers:

 Multi-Head Attention

 Add & Normalize Layer

 Feed-Forward Neural Network (FFN)

 Residual Connections and Dropout ;

Apply Global Average Pooling for dimensionality reduction ;

Stack Multi-Layer Perceptron (MLP) classifier: Dense $\rightarrow$ ReLU $\rightarrow$ Dropout $\rightarrow$ Sigmoid ;

Phase 5: Model Training and Evaluation

Train Transformer model using training data and monitor validation loss ;

Predict labels $\hat{Y}$ on test data ;

Evaluate performance using Accuracy, Precision, Recall, and F1-score ;

Algorithm 1. Network intrusion detection using wrapper-based feature selection and multi-head attention transformer.

The UNSW-NB15 dataset is a network intrusion dataset developed by the Australian Centre for Cyber Security (ACCS) to support the development and evaluation of intrusion detection systems. It is designed to address the limitations of older datasets by incorporating realistic network traffic that includes both benign and malicious activities generated. The dataset consists of 49 features, including basic attributes (such as protocol, source/destination IPs), content features, time-based features as shown in Table 4. The dataset consists of nine types of intrusions including Shellcode, Generic, Worms, Fuzzers, DoS, Reconnaissance, Analysis, Backdoor and Exploits offering a comprehensive benchmark for evaluating machine learning-based network intrusion detection systems.

For experimental setup, the dataset is split into training and testing subsets, with the training set containing 175,341 records and the testing set consisting of 82,332 records. This separation ensures unbiased evaluation of detection models. The rich variety of attacks, combined with realistic traffic simulation, makes UNSW-NB15 a widely used and reliable dataset for benchmarking intrusion detection approaches in contemporary cybersecurity research. To control the model overfitting, we used the validation data to monitor the model at the

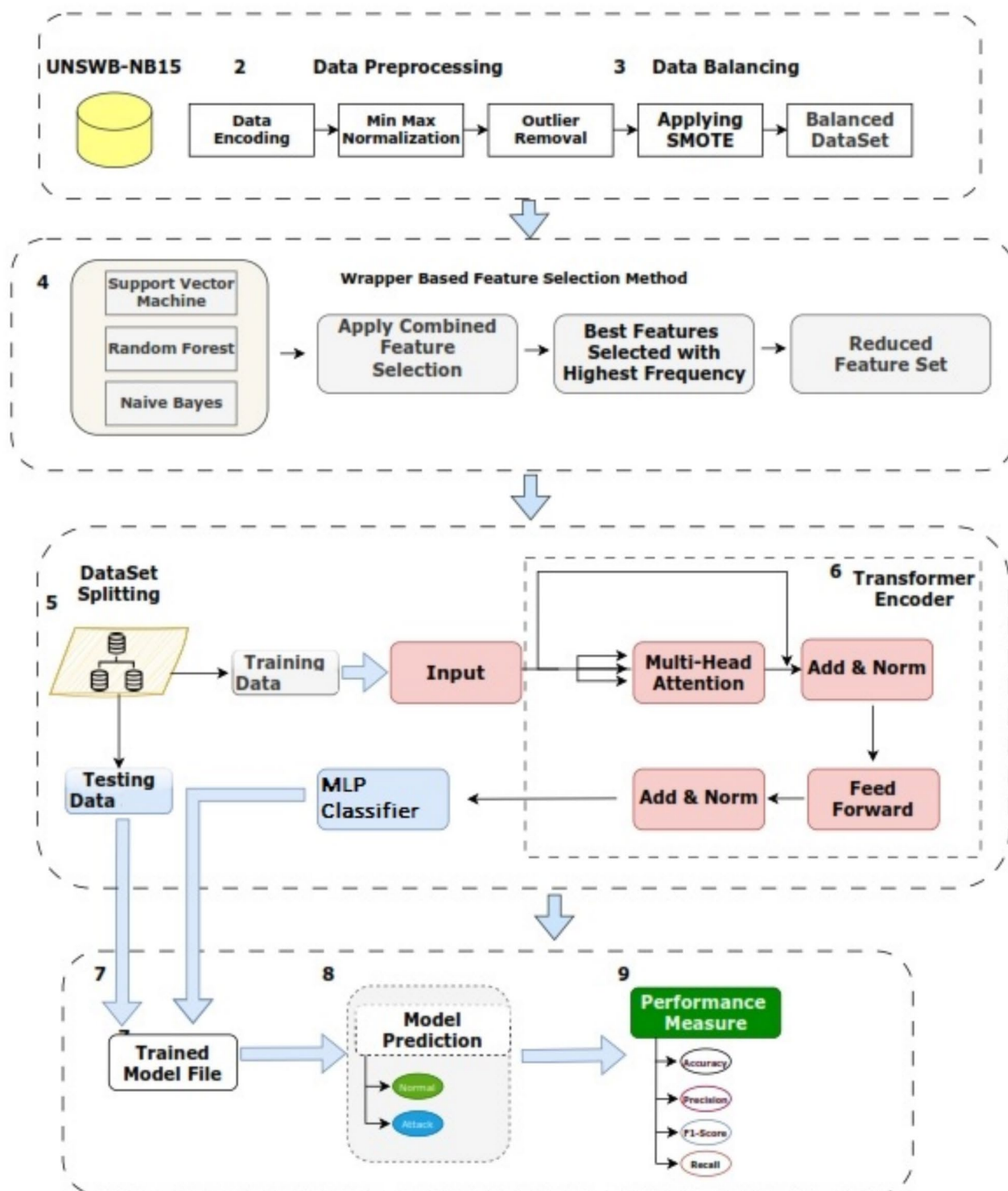


Fig. 2. Architecture of the proposed model.

Technique	Dataset Used	Weaknesses
RTIDS ⁴¹	UNSW-NB15	Ineffective low-dimensional representation through the decoder
Visual transformer with sliding window mechanism and hierarchical focal loss function ⁴²	CICIDS2017, UNSW-NB15	Focal loss did not fully resolve class imbalance issues
Transfer learning-based Transformers ⁴³	NSL-KDD	Complex computational resources are required; Low detection accuracy
Transformers ⁴⁴	CICIDS2017, UNSW-NB15	Use of decoder degrades performance
Transformer model ⁴⁵	CICIDS2017	Longer training time
Improved transformer with stacked autoencoder and KNN-based undersampling with borderline-SMOTE ⁴⁶	NSL-KDD	Class imbalance issues
Visual Transfwormers ⁴⁷	NSL-KDD	Complex computational resources are required
ANN ⁴⁹	UNSW-NB15	No use of feature selection; Low detection accuracy

Table 3. Comparative analysis of transformer-based intrusion detection systems.

time of training. Therefore, we further divided the training data into training and validation data using stratified sampling technique, keeping 90% for training and 10% for validation.

Data encoding

The dataset on intrusion detection consists of character-based features. Since the models can only interpret data consisting of numbers, these values based on characters must be converted into numerical ones. For data encoding Word2Vec method is employed to convert categorical and symbolic features into meaningful numerical vector representations. Traditional encoding techniques like one-hot encoding can result in sparse and high-dimensional vectors, which are inefficient and may not capture semantic relationships between categories. In contrast, Word2Vec, originally designed for natural language processing, learns dense and continuous vector embeddings based on contextual similarity. This approach enables the encoding of features such as service types, protocols, and flags in a way that preserves their underlying relationships, improving the effectiveness of downstream machine learning models by providing richer, context-aware feature representations.

Outlier removal

Data outliers have a negative impact on transformer-based model performance. Outliers are the data points which are far away from the other observations present in the dataset. The reason for the presence of outliers is due to variable data points. Various approaches have been followed by researchers in the past to detect outliers. Z-score is used to detect and remove outliers.

Data normalization

In the data normalization phase, the Min-Max normalization method is applied to scale numerical features into a uniform range, typically [0, 1]. This technique transforms each feature value by subtracting the minimum value of the feature and dividing by the range (i.e., the difference between the maximum and minimum values). Min-Max normalization is essential to ensure that all features contribute equally during model training, especially for algorithms sensitive to the scale of input data such as neural networks. By normalizing the data, it becomes easier for the model to converge during training and avoid bias toward features with larger numerical ranges. This preprocessing step is particularly important in intrusion detection tasks, where balanced feature representation can significantly impact classification performance. The formula is shown in Equation 1.

$$p = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{1}$$

In Equation 1, x is a given value of the data, $x_{\min}$ is the lowest number in the column, $x_{\max}$ is the highest number of that column, and p is the normalized value.

Handling class imbalance problem

Imbalanced datasets, where one class greatly outnumbers others, present a significant challenge for machine learning algorithms. Since normal network traffic often far exceeds actual attacks, models can become biased toward the majority class. To overcome this, authors in⁵¹ proposed a data-generating technique for generating SMOTE, which produces artificial samples to balance the class based on distribution by oversampling the minority class. Since then, SMOTE has been widely used in intrusion detection. The authors in⁵² demonstrated that SMOTE is effective in improving intrusion detection in computer networks and mitigating the effects of class imbalance. In this study, we apply SMOTE to balance the dataset, which improves the detection performance of the model.

Feature selection

This method ranks features in order of the relevance. Techniques for feature selection are divided into two categories: wrapper-based and filter-based. However, filter-based feature selection methods have a straightforward technique and low complexity since they identify the best attributes based on their influence⁵³. All these

Number	Name	Type	No.	Name	Type
1	srcip	Categorical	26	res_bdy_len	Numerical
2	sport	Numerical	27	Sjit	Numerical
3	dstip	Categorical	28	Djit	Numerical
4	dsport	Numerical	29	Stime	Datetime
5	proto	Categorical	30	Ltime	Datetime
6	state	Categorical	31	Sintpkt	Numerical
7	dur	Numerical	32	Dintpkt	Numerical
8	sbytes	Numerical	33	tcprrt	Numerical
9	dbytes	Numerical	34	synack	Numerical
10	sttl	Numerical	35	ackdat	Numerical
11	dttl	Numerical	36	is_sm_ips_ports	Binary
12	sloss	Numerical	37	ct_state_ttl	Numerical
13	dloss	Numerical	38	ct_flw_http_mthd	Numerical
14	service	Categorical	39	is_ftp_login	Binary
15	Sload	Numerical	40	ct_ftp_cmd	Numerical
16	Dload	Numerical	41	ct_srv_src	Numerical
17	Spkts	Numerical	42	ct_srv_dst	Numerical
18	Dpkts	Numerical	43	ct_dst_ltm	Numerical
19	swin	Numerical	44	ct_src_ltm	Numerical
20	dwin	Numerical	45	ct_src_dport_ltm	Numerical
21	stcpb	Numerical	46	ct_dst_sport_ltm	Numerical
22	dtcpb	Numerical	47	ct_dst_src_ltm	Numerical
23	smeansz	Numerical	48	attack_cat	Categorical
24	dmeansz	Numerical	49	label	Binary
25	trans_depth	Numerical			

Table 4. UNSW-B15 features list.

complications can be avoided, and the model can be made more accurate if only the most relevant features are chosen and classified at a later stage. In wrapper-based learning, an algorithm is used that searches for the best characteristics. Different subsets of features are considered, which provides the final classification criterion. The idea of choosing features is to find the best influential attributes and then target them with a reduced dimension that optimizes classification. In this work, we use a method of choosing the best features using three different machine learning algorithms: SVM, RM, and NB. Thee details are discussed below.

Support vector machine

Support Vector Machine (SVM) is a supervised learning approach that applied to situations involving regression and classification. Both linear and non-linear data can be used. When SVM receives N input features from two targets, it uses the predictor values to plot these features graphically in an n -dimensional space or on a two-dimensional axis. The core idea of SVM classification is finding a hyperplane that can classify two distinct classes⁴⁵. By classifying different classes using an N -dimensional hyperplane, SVM is able to handle nonlinear classifications. In this work, a conventional SVM has been chosen for classification. Finding data instances closest to the decision boundary also referred to as support vectors is one of primary computational features of SVM.

Random forest

Random Forest (RF) is an effective machine learning approach primarily used in regression analysis, feature selection, and classification. This technique focuses on the principles of ensemble learning, which combines more than one decision tree (DT) to reduce overfitting problems and improve the prediction score. Each of the decision trees that RF creates is trained using a random part of the input features and training data. The final prediction is generated during prediction by averaging the output from each tree. The randomization helps create distinct and uncorrelated trees, leading to the production of a more dependable and accurate model. In practice, RF may comprise tens of thousands or more decision trees, where each set has a unique set of random characteristics and different subparts of data. In situations where there are many input attributes and a large amount of data, RF can result in extremely accurate and reliable models.

Naïve bayes

It is a well-known learning classifier using the Bayes' theorem. To simplify the estimation of the posterior probability, it makes assumptions about input features that they are conditionally independent, given the output label. Equation 2 shows how Naïve Bayes develops a model $f(x)$ that maximizes the posterior probability given a training set of n instances $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$, where x_i denotes the input attributes and y_i the related output label.

$$f(x) = \arg \max P(y|x) = \arg \max P(x|y) \cdot P(y) \quad (2)$$

Equation 2 uses $P(x|y)$ to represent the likelihood of the input attributes given the output label, and $P(y)$ to represent the prior probability of the output label. To simplify the calculation of $P(x|y)$, this algorithm assumes that the input features are conditionally independent, leading to Equation 3:

$$P(x|y) = \prod_i P(x_i|y) \quad (3)$$

In this case, $P(x_i|y)$ is the likelihood of the i_{th} input feature provided the output label. Equation 4 shows the forecast of the model for a new input x .

$$f(x) = \arg \max P(y) \cdot \prod_i P(x_i|y) \quad (4)$$

In above equation, $\prod_i P(x_i|y)$ is the product of the conditional probabilities of the input attributes given the output label, and $P(y)$ is the prior probability of the output label.

Use of transformers in intrusion detection system

The Transformer was originally developed for natural language processing tasks, which is different from typical RNN and CNN architectures, relying solely on the attention mechanism to achieve machine translation with exceptional outputs. Unlike the sequential models that rely on RNN, all features in this approach can be trained in parallel, significantly improving computational efficiency and reducing time for training. Thus, in this article, we use the transformer as the base model to accelerate feature learning and extraction, thereby improving the speed of training. In contrast to sequence-to-sequence tasks like translation, network intrusion detection tasks do not need decoding steps; instead, a sigmoid activation function is used, and an encoder is employed to learn the most relevant features for classification. Two significant sub-layers of the encoder Transformer, as shown in Figure 3, are feed-forward neural networks and multi-head attention mechanisms, residual and normalization modules. Multi-head attention expands a subspace that captures various bits of information and ultimately improves score of the model by tracking more correlated pieces of information.

As shown in Fig. 3, the model starts with the input layer. The heart of the model is the Transformer encoder, which consists of several important layers to extract and transform information. Using numerous “heads” that investigate different parts of the input data, the multi-head attention layer enables to give importance on multiple portions of the input sequence at once. Each head observes some slightly different relationships in the sequence, helping the model to detect complex patterns. After the attention layer, the data pass through the Add & Norm layer. This layer serves two purposes: residual connection, bringing the original input back into the attention output, and normalizing the data afterward. Residual connections stabilize the training by allowing information to pass through the network more smoothly, whereas normalization keeps the distribution of the data under control. Residual connections also help to avoid the gradient vanishing problem, which is common in deep networks. After attention and normalization, the data pass through a feed-forward network (FFN), a two-layer neural network that further refines the learned representation. The first FFN uses ReLU activation as a nonlinear layer and returns the output to the original embedding dimension. The FFN layers apply these transformations independently to each token, which enhances the features learned by the attention layer and enables the model to learn more complex patterns. After the encoder, Global Average Pooling is applied to decrease the dimensionality of the data by averaging across the time dimension of the sequence. This produces a vector of fixed size for any input and simplifies the application to classification problems. To prevent overfitting, dropout layers are used, where a portion of the inputs is randomly set to zero during the training phase. This prevents the model from becoming overly reliant on specific input patterns. Dropout is important for avoiding overfitting and to make sure that the model generalizes well to unseen data. Finally, the pre-processed data is given to a layer with 20 units, then by ReLU activation and a sigmoid layer for classification.

The Multi-Layer Perceptron (MLP) classifier is used that consists of three layers i.e., two hidden layers and one output layer. The hidden layers utilize the ReLU activation function, which introduces non-linearity and helps the network learn complex patterns in the data. To prevent overfitting and improve generalization, a dropout mechanism with a ratio of 0.2 is applied to these hidden layers, randomly deactivating 20% of the neurons during training. The final layer, an output layer, uses the sigmoid activation function, making the model suitable for binary classification tasks by producing output values between 0 and 1 that can be interpreted as probabilities. This architecture balances expressiveness and regularization, improving the ability of the model to make accurate predictions.

Performance evaluation metrics

Different metrics are used for evaluating the performance of the proposed model. These metrics are described below.

- **Accuracy** It is the proportion of samples that were correctly classified out of all samples. It is a useful measure when the dataset is balanced. However, since normal samples are significantly more common than anomalous samples in real network contexts, accuracy may not be an appropriate measure. Equation 5 is used to calculate accuracy.

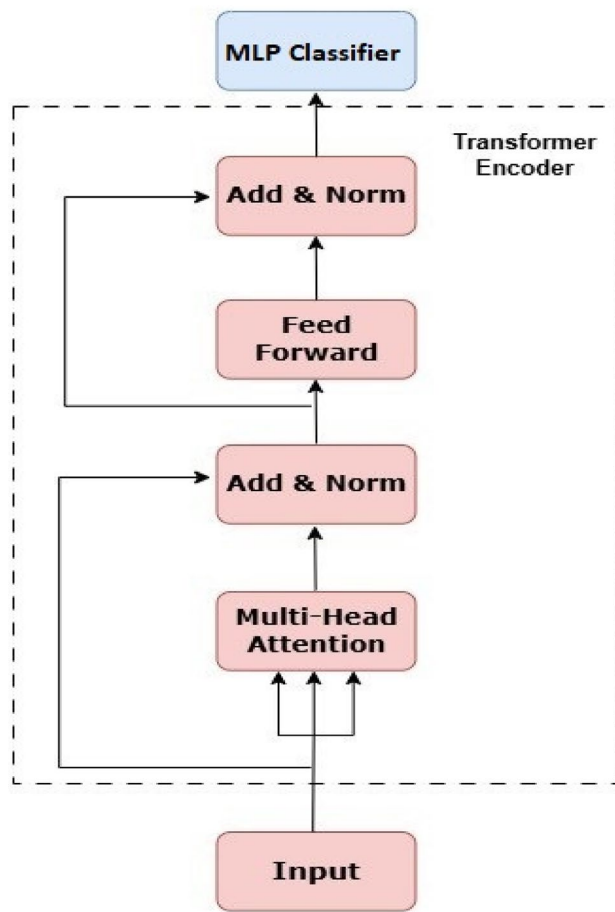


Fig. 3. Transformer encoder block.

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN} \quad (5)$$

Where TP is true positive, TN is true negative, FP is false positive, and FN is false negative.

- **Precision** Precision measures how many positive classifications are actually true positives. Precision is calculated using Equation 6:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (6)$$

- **Recall** Recall evaluates how many actual positive instances were identified by the model. It is calculated using Equation 7:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (7)$$

- **F1-Score** F1-Score is the harmonic mean of precision and recall and provides a balance between them. It is calculated using Equation 8:

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

Results and discussion

This section presents the evaluation of the proposed model on the UNSW-NB15 dataset. The analysis of different results obtained is also presented. The results are organized according to the stated problem and research objectives.

Experimental setup

To evaluate the proposed network intrusion detection model, we used Google Colab as the computational platform. The model was implemented using Python with libraries such as TensorFlow, Keras, Scikit-learn, and Pandas.

Multiple experiments were conducted using different methods for balancing data. The purpose of these experiments was to assess which model performs best in handling class imbalance issues. Three different sampling methods were considered i.e., random oversampling, random undersampling, and the use of SMOTE. The results are given in Table 5.

The results shown in Table 5 demonstrate how well various sampling techniques improve classification metrics. SMOTE outperforms both Random Oversampling and Random Undersampling in terms of accuracy, precision, recall, and F1-score, getting a score of 92.7% and an F1-score of 92%. This aligns with findings from the literature, where other sampling methods are used to address class imbalance and overlap issues to improve model performance.

Random Oversampling achieves a score of 91.5%, which is less than SMOTE. This method samples data points randomly, which can sometimes lead to over-representation of noise in the dataset, limiting its ability to fully resolve class imbalance. Random Undersampling shows the lowest accuracy (88.3%) among the three methods. While it reduces dataset size and computational cost, it may discard useful samples, which can degrade model performance, particularly in complex datasets. The superiority of SMOTE in this comparison can be attributed to its balanced approach of generating synthetic samples based on the feature space of minority classes. Unlike random methods, SMOTE avoids over-representing noise and instead enhances class separability, thereby achieving better precision, recall, and F1-score. This analysis highlights that advanced oversampling methods like SMOTE provide a more robust solution for addressing class imbalance, leading to improved classification outcomes across key metrics.

Comparison experiments with and without feature selection

Experiments were conducted to assess the performance of the model with and without the feature selection method. The goal was to evaluate how well the proposed model performs with no feature selection versus when the wrapper-based feature selection method is used. The results are shown in Table 6.

The table highlights the importance of using feature selection for this model. When comparing the two approaches, it is evident that using a feature selection method improves all key evaluation metrics, including accuracy, precision, recall, and F1-score. Without feature selection, the model achieves a score of 0.90, with precision of 0.88, recall of 0.89, and F1-score of 0.88. The inclusion of all features likely introduces noise or irrelevant information, which may restrict the ability of the model to generalize well, though the results still show a reasonably high model performance. In contrast, using feature selection improves accuracy to 93%, with precision, recall, and F1-score all increasing to 0.91, 0.92, and 0.92, respectively. Feature selection improves model performance by retaining only the most relevant attributes, minimizing dimensionality and mitigating the effects of noisy or redundant data.

Comparative experiments with and without multi-head attention network in transformers

Experiments were conducted to find out the working of the proposed model with and without the use of multi-head attention in the transformer model. The goal was to determine if the performance of the model gets better due to the use of multi-head attention. The results of these experiments are shown in Table 7.

The results show that incorporating multi-head attention significantly improves the performance of the model across all metrics. With multi-head attention, the model achieves a detection score of 0.92, precision of 0.90, recall of 0.91, and an F1-score of 0.91. In contrast, without multi-head attention, the model attains a detection score of 0.87, precision of 0.88, recall of 0.85, and an F1-score of 0.87. This improvement highlights the power of multi-head attention in enhancing the ability of the model to identify difficult patterns and connections in the data.

Comparison with recent intrusion detection models

To assess the detection capabilities of the proposed model, this study compares it against several recently proposed intrusion detection models that have demonstrated strong performance on the UNSW-NB15 dataset. These include CNN⁵⁴, CNN-LSTM⁵⁴, CBA-CLSVE⁵⁵, SSC-OCSVM⁵⁶, CNN-GRU⁵⁷, and FCNN-SE⁵⁸. Additionally, the study includes intrusion detection models based on Transformers, such as RTIDS⁵⁶, VIT⁵⁹, and CNN-Transformer⁵⁹, for further comparison. The experimental results show the robustness of the proposed model, which outperforms all others in detecting intrusions. The results are shown in Table 8 and Fig. 4.

The Table presents a detailed comparison of various models used for a classification task. Among all the approaches, the proposed model stands out with the highest performance, achieving a detection score of 93%, precision of 91%, recall of 92%, and F1-score of 92%. These results concludes that the proposed model

Sampling Methods	Accuracy	Precision	Recall	F1-score
Random Oversampling	0.91	0.89	0.91	0.90
Random Undersampling	0.88	0.85	0.87	0.86
SMOTE	0.92	0.91	0.93	0.92

Table 5. Comparison of different methods for balancing datasets.

Technique	Accuracy	Precision	Recall	F1-score
Without Using Feature Selection	0.90	0.88	0.89	0.88
By Using Feature Selection	0.93	0.91	0.92	0.92

Table 6. Comparison of using feature selection technique.

Technique	Accuracy	Precision	Recall	F1-score
Without Multi-Head Attention	0.87	0.88	0.85	0.87
With Multi-Head Attention	0.92	0.90	0.91	0.91

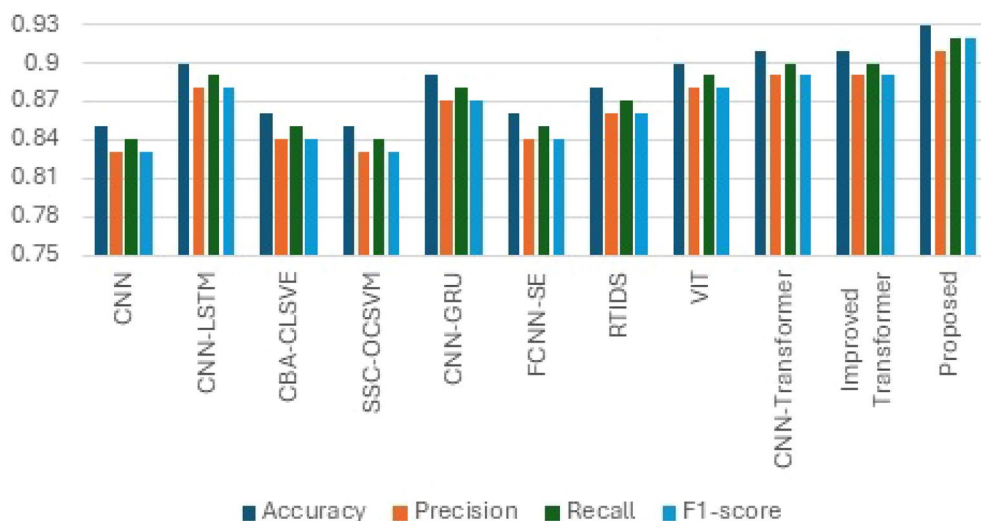
Table 7. Comparison of performance with and without multi-head attention.

Model	Accuracy	Precision	Recall	F1-score
CNN ⁵⁴	0.85	0.83	0.84	0.83
CNN-LSTM ⁵⁴	0.90	0.88	0.89	0.88
CBA-CLSVE ⁵⁵	0.86	0.84	0.85	0.84
SSC-OCSVM ⁵⁶	0.85	0.83	0.84	0.83
CNN-GRU ⁵⁷	0.89	0.87	0.88	0.87
FCNN-SE ⁵⁸	0.86	0.84	0.85	0.84
RTIDS ⁵⁶	0.88	0.86	0.87	0.86
VIT ⁵⁹	0.90	0.88	0.89	0.88
CNN-Transformer ⁵⁹	0.91	0.89	0.90	0.89
Improved Transformer ⁵⁹	0.91	0.89	0.90	0.89
Proposed	0.93	0.91	0.92	0.92

Table 8. Comparison of existing methods with proposed model on UNSW-NB15 dataset.

incorporates advanced techniques or optimizations that allow it to effectively handle the dataset's complexities and deliver superior predictions. On the lower end of the performance spectrum, models such as CNN, CBA-CLSVE, and SSC-OCSVM demonstrate accuracies ranging from 85%-86%. These models rely on simpler architectures or methods that struggle with challenges like class imbalance and feature redundancy, resulting in relatively lower overall performance.

The higher performing models, including CNN-LSTM, VIT, CNN-Transformer, and improved transformer, achieve accuracies of 90%-91%. These architectures combine traditional convolutional networks with advanced techniques like transformers, enabling them to identify intricate relationships in the data. Despite their strong results, they are surpassed by the proposed approach, which likely benefits from further refinements such

**Fig. 4.** Comparative results of the proposed model with existing methods.

as feature selection, improved pre-processing, or specialized model layers tailored to the task. The superior performance of the proposed model can be attributed to the use of preprocessing steps like outlier removal, min-max normalization, feature scaling, wrapper-based feature selection to choose the best 20 features, and the utilization of multi-head attention.

Conclusion

This article presents an optimized transformer-based framework. A wrapper-based feature selection technique is used to ensure that only the most relevant attributes are used. SMOTE is used to effectively resolve the issues of data imbalance, resulting in improved metrics across accuracy, precision, recall, and F1-score. The proposed model uses the integration of the multi-head attention mechanism, which allows the transformer to focus on multiple aspects of input data simultaneously. Additionally, the use of dimensionality reduction techniques and dropout layers prevents overfitting and ensures robust performance. The proposed model achieves an accuracy of 93%, precision of 91%, recall of 92%, and an F1-score of 92% on the UNSW-NB15 dataset, highlighting its effectiveness in detecting intrusions. The results demonstrate the potential of combining advanced feature selection with Transformer-based architecture for tackling complex security challenges.

While the multi-head attention mechanism has proven effective, integrating more advanced techniques, such as sparse or adaptive attention, may improve the capability of the model to focus on critical attributes while reducing computational complexity. Hybrid approaches, combining Transformers with other techniques like generative adversarial networks (GANs) or ensemble methods, might offer further performance enhancements.

Data availability

The datasets used in this article are publicly available on <https://research.unsw.edu.au/projects/unsw-nb15-dataset>

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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